**Read Carefully!**

Name: _____

**Computer Science KNN Project**

1. (5 points)
   Use the dataset on the other page to complete the following project.

   - Modify the KNN code in class to predict if a student is a lower classmen (freshman/sophomore) or upper classmen (junior/senior).
   - You will have 5 students to predict, and need to make a prediction for $k = 1, 3, 5$. There must be 15 total answers.
   - I only need one code example. You can comment the answers in your program.

Table 1: KNN predictions

| Student | AP | Honors | Clubs | Screen Time | Sports Seasons | Class |
|---|---|---|---|---|---|---|
| A | 9 | 5 | 3 | 9 | 5 | Junior |
| B | 2 | 3 | 1 | 6.0 | 10 | Senior |
| C | 10 | 3 | 2 | 5.5 | 8 | Senior |
| D | 9 | 4 | 4 | 4.5 | 12 | Senior |
| E | 3 | 2 | 5 | 6.0 | 10 | Senior |

Here is the code from class:

```
import math

# [age, education, salary, sex]
# education: 1=HS, 2=BA, 3=MA, 4=PhD
# sex: 0=female, 1=male

people = [
    ([22, 1, 30000, 0], "Democrat"),
    ([25, 2, 40000, 1], "Democrat"),
    ([31, 2, 50000, 0], "Democrat"),
    ([35, 3, 65000, 1], "Democrat"),
    ([28, 2, 45000, 0], "Democrat"),
    ([42, 3, 80000, 1], "Democrat"),

    ([23, 1, 35000, 1], "Republican"),
    ([30, 2, 55000, 1], "Republican"),
    ([45, 3, 75000, 0], "Republican"),
    ([60, 4, 120000, 1], "Republican")
]

# New person
new_person = [40, 3, 70000, 1]


# Calculate Euclidean distance
def distance(person1, person2):
    total = 0

    for i in range(len(person1)):
        total += (person1[i] - person2[i]) ** 2

    return math.sqrt(total)
```

```
33
34
35 # Calculate distances from new person
36 distances = []
37
38 for person, label in people:
39     d = distance(new_person, person)
40     distances.append((d, label, person))
41
42
43 # Sort from closest to farthest
44 distances.sort()
45
46
47 # Look at the 3 nearest neighbors
48 k = 3
49
50 print("Nearest neighbors:")
51
52 for d, label, person in distances[:k]:
53     print(label, "distance =", round(d, 2))
54
55
56 # Vote
57 democrats = 0
58 republicans = 0
59
60 for d, label, person in distances[:k]:
61     if label == "Democrat":
62         democrats += 1
63     else:
64         republicans += 1
65
66
67 if democrats > republicans:
68     prediction = "Democrat"
69 else:
70     prediction = "Republican"
71
72
73 print("\nPrediction:", prediction)
```

Here is the dataset to train from:

*I pledge that I have upheld both the letter and the spirit of the Baylor Honor Code, neither giving nor receiving unauthorized assistance on this assessment.*

Signed: _____

Table 2: KNN Student Dataset

| Student | AP | Honors | Clubs | Screen Time | Sports Seasons | Class |
|---|---|---|---|---|---|---|
| 1 | 0 | 2 | 2 | 5.5 | 2 | Freshman |
| 2 | 0 | 1 | 1 | 6.5 | 1 | Freshman |
| 3 | 1 | 2 | 3 | 4.5 | 3 | Freshman |
| 4 | 0 | 3 | 2 | 7.0 | 2 | Freshman |
| 5 | 1 | 1 | 4 | 5.0 | 1 | Freshman |
| 6 | 0 | 2 | 1 | 6.0 | 4 | Freshman |
| 7 | 1 | 3 | 2 | 4.0 | 2 | Freshman |
| 8 | 0 | 2 | 4 | 5.5 | 3 | Freshman |
| 9 | 1 | 2 | 1 | 7.0 | 1 | Freshman |
| 10 | 0 | 1 | 3 | 6.0 | 2 | Freshman |
| 11 | 1 | 3 | 3 | 5.0 | 3 | Sophomore |
| 12 | 1 | 2 | 4 | 4.5 | 4 | Sophomore |
| 13 | 2 | 2 | 2 | 6.0 | 3 | Sophomore |
| 14 | 1 | 4 | 3 | 5.5 | 2 | Sophomore |
| 15 | 2 | 3 | 5 | 4.0 | 4 | Sophomore |
| 16 | 1 | 2 | 2 | 7.0 | 3 | Sophomore |
| 17 | 2 | 4 | 4 | 5.0 | 5 | Sophomore |
| 18 | 1 | 3 | 5 | 6.0 | 2 | Sophomore |
| 19 | 2 | 2 | 3 | 5.5 | 4 | Sophomore |
| 20 | 1 | 4 | 2 | 4.5 | 3 | Sophomore |
| 21 | 2 | 3 | 4 | 4.5 | 5 | Junior |
| 22 | 3 | 2 | 3 | 5.5 | 4 | Junior |
| 23 | 2 | 4 | 5 | 4.0 | 5 | Junior |
| 24 | 3 | 3 | 2 | 6.0 | 3 | Junior |
| 25 | 2 | 5 | 4 | 4.5 | 6 | Junior |
| 26 | 3 | 3 | 6 | 5.0 | 4 | Junior |
| 27 | 2 | 4 | 3 | 6.5 | 5 | Junior |
| 28 | 4 | 2 | 5 | 4.0 | 6 | Junior |
| 29 | 3 | 4 | 4 | 5.5 | 5 | Junior |
| 30 | 2 | 3 | 6 | 4.5 | 4 | Junior |
| 31 | 3 | 4 | 5 | 4.0 | 5 | Senior |
| 32 | 4 | 3 | 4 | 5.0 | 6 | Senior |
| 33 | 3 | 5 | 6 | 3.5 | 6 | Senior |
| 34 | 5 | 3 | 5 | 4.5 | 7 | Senior |
| 35 | 4 | 4 | 7 | 4.0 | 6 | Senior |
| 36 | 3 | 2 | 4 | 6.0 | 5 | Senior |
| 37 | 5 | 4 | 6 | 3.5 | 8 | Senior |
| 38 | 4 | 5 | 5 | 5.0 | 7 | Senior |
| 39 | 3 | 3 | 7 | 4.5 | 6 | Senior |
| 40 | 4 | 4 | 6 | 5.5 | 5 | Senior |